Saragadam Lokesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.16
Accent junior college	Class XII (SSE)	2023	92.50%
Sri Sathya Sai Vidhya Vihar	Class X (CBSE)	2021	89.00%

INTERNSHIP EXPERIENCE

Accelequant India Pvt. Ltd. -- Quantum Engineering Intern

Bangalore | Jan 2026 - Jul 2026

- Developed and benchmarked **hybrid quantum-classical optimization pipelines** for Linear Programming (LP) and Mixed-Integer Linear Programming (MILP) problems using **Python, Qiskit and Gurobi**.
- Implemented and evaluated **constraint-aware VQA and quantum-improved local-search approaches**, addressing variable bounds, feasible-group construction, constraint handling and iterative parameter updates.
- Investigated **convergence, reproducibility and solution quality** across optimization formulations, comparing quantum/hybrid approaches against classical solvers and heuristics.
- Built preprocessing and scaling workflows for optimization problems and evaluated algorithm behavior on increasingly realistic problem formulations.

PROJECTS

1.CodeTrack -- Full-Stack DSA Progress & Analytics Platform | Github-Repo | Project-Link

- Built a full-stack DSA analytics platform integrating **LeetCode and Codeforces** activity, ratings, submissions and topic-wise performance using React and Node.js/Express.
- Designed **PostgreSQL/Prisma schemas and REST APIs** with JWT authentication, bcrypt password hashing, protected routes and submission-state tracking for Accepted, Wrong Answer, TLE and MLE outcomes.
- Developed a **Chrome extension-based submission ingestion pipeline** to overcome platform API limitations, handling duplicate requests and database upserts.
- **Deployed the application as a distributed web stack**, hosting the PostgreSQL database on **Supabase**, backend services on **Render**, and the React frontend on **Vercel**, with cross-service API integration and CORS configuration.
- Built an **operations/observability dashboard** with request history, filtering, pagination and event inspection, backed by a bounded **1,000-event in-memory event store** and request-ID correlation for failure diagnostics.
- Implemented production-oriented **security and reliability controls**, including server-side admin authorization, log sanitization, CSP, request timeouts/abort handling, controlled polling and automated backend tests.

Tech: React, Node.js, Express, PostgreSQL, Prisma, JWT, Chrome Extension

2.Personal Knowledge Assistant | RAG + LLM | Github-Repo

- Built a retrieval-augmented generation pipeline for question answering over a curated collection of documents using **FAISS, embeddings and Gemini**.
- Implemented document chunking and metadata-aware retrieval to preserve **source/page information** for retrieved passages.
- Developed a Streamlit interface for interactive question answering with **source-aware responses and citation tracking**.
- Built an evaluation workflow to assess **answer quality, retrieval/source quality and citation quality**, identifying retrieval failures and opportunities for improving context selection.

Tech: Python, FAISS, Gemini, Streamlit, RAG, Embeddings

SKILLS

- **Languages:** Python, C++, C, SQL, JavaScript
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, RAG, FAISS

- **Backend:** Node.js, Express.js, PostgreSQL, Prisma, REST APIs, JWT
- **Tools/Cloud:** Git, GitHub, Qiskit, Gurobi
- **Frameworks:** React, Streamlit

EXTRACURRICULAR

- **Silver Medal, Milan Cultural Fest, IIT Hyderabad (2025)** -- 2nd Place, Team Cosplay
- **Codeforces:** 1230 rating | **Chess:** District-level tournament participant

RELEVANT COURSES

- **Relevant Courses:** Artificial Intelligence (AI/ML), Numerical Methods, Linear Algebra, Differential Equations, Applied Mathematics,Intro to Programming,Data Structures and Applications.

ACHIEVEMENTS

- **Secured 9.83 SGPA (Semester 3)** and **9.73 SGPA (Semester 4)** -- among top performers in the department.